For physicists' current understanding of backward time travel in four spacetime dimensions without a bulk, see the last chapter of *Black Holes & Time Warps* (Thorne 1994), the chapters by Hawking, Novikov, and me in *The Future of Spacetime* (Hawking et al. 2002), and *Time Travel and Warp Drives* (Everett and Roman 2012). These are all by physicists who have contributed in major ways to the theory of time travel. For a historical account of modern research on time travel, see *The New Time Travelers: A Journey to the Frontiers of Physics* (Toomey 2007). For a comprehensive discussion of time travel in physics, in metaphysics, and in science fiction, see *Time Machines: Time Travel in Physics, Metaphysics and Science Fiction* (Nahin 1999). *From Eternity to Here: The Quest for The Ultimate Theory of Time* (Carroll 2011) is a wonderful discussion of almost everything physicists know, or speculate, about the nature of time.

I don't know any good books or articles, for general readers, about time travel when our universe is a brane that lives in a higher dimensional bulk; but as I discuss in Chapter 30, Einstein's laws extended to higher dimensions give basically the same predictions as without a bulk.

For some technical details of Cooper's sending messages backward in time to Murph, see the appendix *Some Technical Notes*.

Chapter 31. Lifting Colonies off Earth

For Murph's method (reducing G) for lifting the colonies off Earth, in my interpretation of *Interstellar,* see my remarks about Chapter 25, above.

In the early 1960s, when I was a PhD student at Princeton University, one of my physics professors, Gerard K. O'Neill, was embarking on an ambitious feasibility study for colonies in space,